



UTM – FC DIGITAL FORENSIC
UNIVERSITI TEKNOLOGI MALAYSIA

DIGITAL FORENSIC (MECR1033)

LAB 1

*Understanding Evidence File Meta Information
with Exiftool Software*



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SECTION : 52

Instruction: Complete all the following practises.

SECTION A: Understand *Exiftool* software

- 1) **ExifTool** is a command line tool that is used for reading, writing and editing metadata information in photos, audio and videos. It has a wide range of file type support like EXIF, JFIF, GPS, XMP, ICC profile, FlashPix, ID3, Photoshop IRB and more.
- 2) It also recognizes maker notes from many digital cameras like those from Sony, Canon, Casio, Ricoh, Samsung, Pentax/Asahi, Nikon, Nintendo, Motorola, Olympus, Panasonic, Leica, Fujifilm, Minolta, Kodak, HP, Sanyo, Sigma and many more devices.
- 3) It also supports geotag images, generates track logs, renames files, can delete metadata information in groups and individually, automatically backs up the original image, recognizes thousands of tags and more.

SECTION B: Download the *Exiftool* software

- 1) Go to <https://exiftool.org/> , then download and install the Exiftool software
- 2) **OR**
 - a. use online version **ezgif** (<https://ezgif.com/view-metadata>) or
 - b. use online **Omnifile** (<https://omnifile.co/exif/>)
- 3) Kindly to download all the files in the Forensic Lab 1 Instruction
 - a. Image iasrg A.png
 - b. Image iasrg B.png
 - c. Test 1 File_Exitfool.docx
 - d. Test 1a File_Exitfool.docx
 - e. Test 1 b File_Exitfool.docx

SECTION C: Investigate the meta information of the “Test 1 File_Exitfool.doc” file

- 1) Open C Command Prompt, go to the “C:\Windows>” directory
- 2) Type C:\Windows>Exiftool then add one space
- 3) Drag the “**Test 1 File_Exitfool.doc**” next to the word “C:\Windows>Exiftool “, it will then become **C:\Windows>Exiftool xxx\Test 1 File_Exitfool.docx** , then click Enter
- 4) Now, observe the meta information of the file. Explain your result.
- 5) **OR**
 - a. use Online metadata2go (<https://www.metadata2go.com>)
- 6) Alternatively, you may extract the folder at Desktop, make sure the icon for Exitftool has change to  **exiftool(-k)**

- 7) Drag the **"Test 1 File_Exiftool.doc"** to the icon as per example
- 8) Analyze the result showed

Result and Explanation:

Based on the metadata analysis from metadata2go.com, the "Test 1 File_Exiftool.docx" file reveals the following key information:

- File Identity: The file is a DOCX (Microsoft Word) document.
- Authoring Information:
 - The creator and author listed is "Windows User".
 - The last person to modify the file was "Siti Hajar Othman".
- Timeline:
 - File Creation Date: 2020:02:24 03:42:00Z
 - File Modification Date: 2024:10:07 06:50:19Z
 - Total Editing Time: 2 minutes.
- Document Statistics: The document contains 1 page, 19 words, 111 characters, and 1 paragraph.
- Software Used: The file was created or modified using WPS Office (version 12.2.0.18283), an alternative to Microsoft Office.
- File Integrity: The checksum (MD5) is 102468fee024337ebb7405b0eb3a5d6a, which serves as a unique digital fingerprint for the file in its current state.
- Revision Number: The document has been through 2 revision cycles.

In summary, the metadata shows a document initially created by a generic "Windows User" in 2020, which was later edited by Siti Hajar Othman in 2024 using WPS Office, with a total of 2 minutes spent editing it.

SECTION D: Compare the "Test 1 File_Exiftool.doc" and the "Test 1a File_Exiftool.doc"

- 1) Put side-by-side the **"Test 1 File_Exiftool.doc"** and the **"Test 1a File_Exiftool.doc"** exiftool black windows
- 2) Explain the differences you can find from both files

A comparative analysis of the metadata for "Test 1 File_Exiftool.docx" and "Test 1a File_Exiftool.docx" reveals that they are distinct files with different digital histories, despite containing visually identical text content. The most definitive evidence of their uniqueness is their differing checksums 102468fee024337ebb7405b0eb3a5d6a for Test 1 and 3ee0020c14c81ac53b504932a0d494a2 for Test 1a, which act as unique digital fingerprints. The metadata further indicates that "Test 1a" is a later or more edited version, as it has a higher revision number 3 compared to 2 and was created four minutes after the original file (04:25:00Z vs. 03:42:00Z). Additional minor discrepancies include the total editing time, where "Test 1" records 2 minutes while "Test 1a" shows 0, and the character count, which is 111 in "Test 1" and 110 in "Test 1a," confirming a slight content alteration. These metadata differences, including distinct internal identifiers (Icv) and raw header data, conclusively show that "Test 1a" is a separate instance of the file that was saved again, likely resulting in the unnoticeable change in character count that differentiates it from the original "Test 1" file.

3) Then, identify the different on the content of the file

The differences that I find is that Test 1 File_Exiftool.docx contains two paragraphs of text while Test 1a File_Exiftool.docx contains two paragraphs of text. The content is identical to "Test 1" based on the provided file contents. The metadata differences are like revision number and checksum suggest that "Test 1a" is likely a later, re-saved version of the same content, even though the visible text is the same.

SECTION E: Investigate the "iasrg A" and "iasrg B" image files

1) Explain what are the differences of their meta information.

The primary difference lies in their file format and inherent properties, which would be evident in their metadata.

- File Format: "iasrg A" is a PNG file, while "iasrg B" is a JPEG file.
- Metadata Capacity: PNG files are primarily for storing image data and can contain text chunks for metadata. JPEG files have a more robust and standardized structure for embedding EXIF, IPTC, and XMP metadata.
- Compression: JPEG uses lossy compression, which can reduce file size but also quality. PNG uses lossless compression. This would be reflected in the metadata under compression tags.

	Similarity 96.94%		Method Structural Similarity Index Measure	
		First File	Second File	
	Name	Image iasrg B.jpg	Image iasrg A.png	
	Size	12.25 KB	5.53 KB	
	Height	225px	225px	
	Width	225px	225px	
	File Type	jpeg	png	

2) Is there any different in both file's content?

Based on the provided text content for both images, there is no difference in the visible content. Both display the text "Information Assurance and Security Research Group". The difference is purely in the file format and the associated metadata structure.

SECTION F: Documentation: a) Fill up the Digital Evidence Form on your Exiftool analysis

a) Fill up the Digital Evidence Form for your Exiftool analysis

- 1) Create / Download the **Digital Evidence Form** template (could find from internet its sample)
- 2) Select another **5 more additional files** (in any form *.doc, *.jpeg, *.xls, *.pdf etc.) and analyse those files.
- 3) Fill up the form with your investigation analysis

END_OF_LAB1